

Static Gesture Classification — Run Report

Generated 2026-05-14 13:11:18

Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic · 35 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic · 35 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic · 36 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic · 35 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic · 35 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic · 24 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic · 35 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic · 35 trials
Training data path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic · 36 trials
Training data path 10	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic · 35 trials
Training data path 11	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Dynamic · 36 trials
Training data path 12	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic · 35 trials
Training data path 13	/home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic · 35 trials
Total training paths	13 · 447 trials total
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/StaticReports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	176 channels
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats+fft → feature dim 9504
Split / seed / CV	random_state=42 · CV: LOUO (13 folds, every user held out once)
Sample counts	train (post-aug)=1341 · LOUO-evaluated samples=1341

Augmentation (training only)

Enabled	yes
Copies per train trial	2
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — LOUO cross-validation

Algorithm	CV mean (LOUO, 13 folds)	CV std	CV min	CV max
Random Forest	0.8988	0.0805	0.7238	1.0000
Voting Soft (SVM+RF+LogReg)	0.8970	0.0887	0.6571	0.9722
Stacking (SVM+RF+LogReg)	0.8954	0.0901	0.6667	1.0000
Logistic Regression	0.8947	0.0863	0.6571	0.9722
SVM (RBF)	0.8840	0.0828	0.6571	0.9722
SVM (Linear)	0.8809	0.0961	0.6286	0.9722
KNN	0.8253	0.0508	0.7143	0.9167
LDA	0.4775	0.1661	0.2286	0.8000

Per-class CV F1 (across all LOUO folds)

Class	Random Forest	Voting Soft (SVM + RF + LogReg)	Stacking (SVM + RF + LogReg)	Logistic Regression	SVM (RBF)	SVM (Linear)	KNN	LDA	Support
Double_Lower	0.946	0.932	0.902	0.931	0.892	0.888	0.860	0.381	195
Double_Nothing	0.674	0.669	0.714	0.692	0.672	0.683	0.372	0.253	186
Double_Pistol_Recoil	0.886	0.910	0.919	0.880	0.921	0.891	0.797	0.575	195
Double_Raise	0.953	0.948	0.955	0.948	0.953	0.948	0.950	0.722	195
Double_Wiggle	0.977	0.934	0.960	0.939	0.948	0.928	0.885	0.399	192
Double_cmere	0.970	0.985	0.985	0.977	0.977	0.977	0.935	0.653	198
Drum_Roll	0.857	0.845	0.796	0.840	0.800	0.797	0.789	0.394	180

macro avg	0.895	0.889	0.890	0.887	0.880	0.873	0.798	0.482	1341
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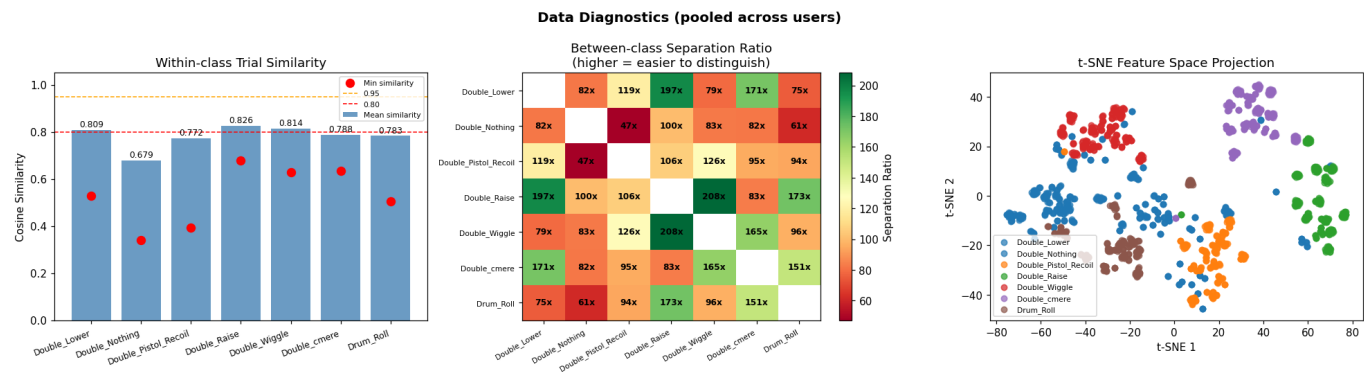
Per-user accuracy (each user as held-out fold)

User	n trials	Random Forest	Voting Soft (SVM + RF + LogReg)	Stacking (SVM + RF + LogReg)	Logistic Regression	SVM (RBF)	SVM (Linear)	KNN	LDA
1_Alan	105	0.962	0.924	0.857	0.895	0.829	0.771	0.800	0.276
2_Alex	105	0.914	0.971	0.943	0.971	0.971	0.971	0.829	0.543
3_Anghad	108	1.000	0.917	0.972	0.917	0.917	0.917	0.833	0.500
4_Daniel	105	0.867	0.914	0.914	0.914	0.914	0.914	0.857	0.486
6_Harry	105	0.952	0.943	0.905	0.943	0.943	0.943	0.829	0.800
8_Jestin	72	0.917	0.958	0.958	0.958	0.875	0.958	0.889	0.458
11_Mansh	105	0.781	0.771	0.771	0.771	0.819	0.800	0.857	0.495
12_Marcus	105	0.724	0.657	0.667	0.657	0.657	0.629	0.714	0.229
14_StephenV2	108	0.944	0.972	0.972	0.972	0.972	0.972	0.917	0.648
15_Tash	105	0.952	0.943	0.962	0.943	0.905	0.886	0.800	0.333
16_Bella	108	0.861	0.861	0.861	0.889	0.861	0.861	0.833	0.639
17_Raquel	105	0.819	0.857	0.857	0.857	0.857	0.857	0.752	0.571
18_Bridgette	105	0.990	0.971	1.000	0.943	0.971	0.971	0.819	0.229
Mean (across users)		0.899	0.897	0.895	0.895	0.884	0.881	0.825	0.477
Std (across users)		0.081	0.089	0.090	0.086	0.083	0.096	0.051	0.166

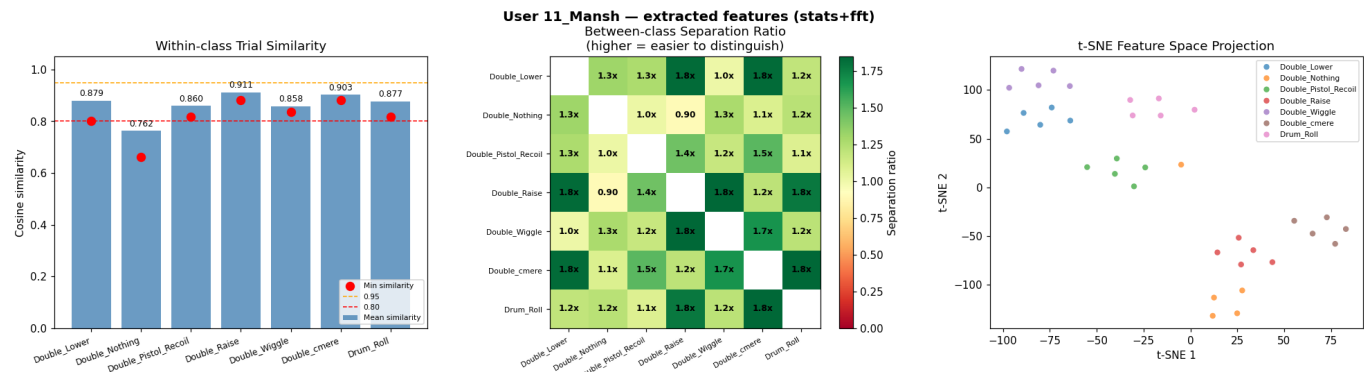
Appendix — Data Diagnostics

14 figures: pooled across users + one per detected user (extracted-feature views).

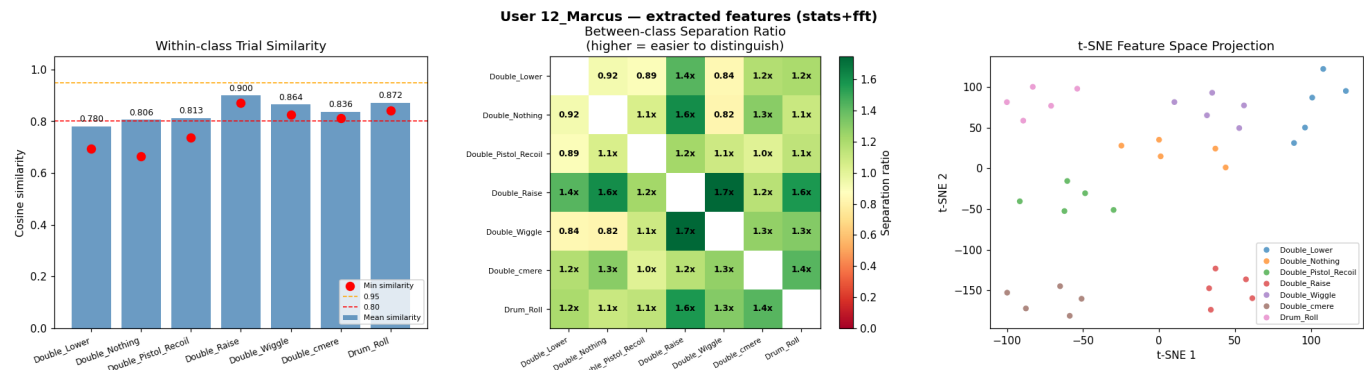
Pooled diagnostics — extracted features (across all users)



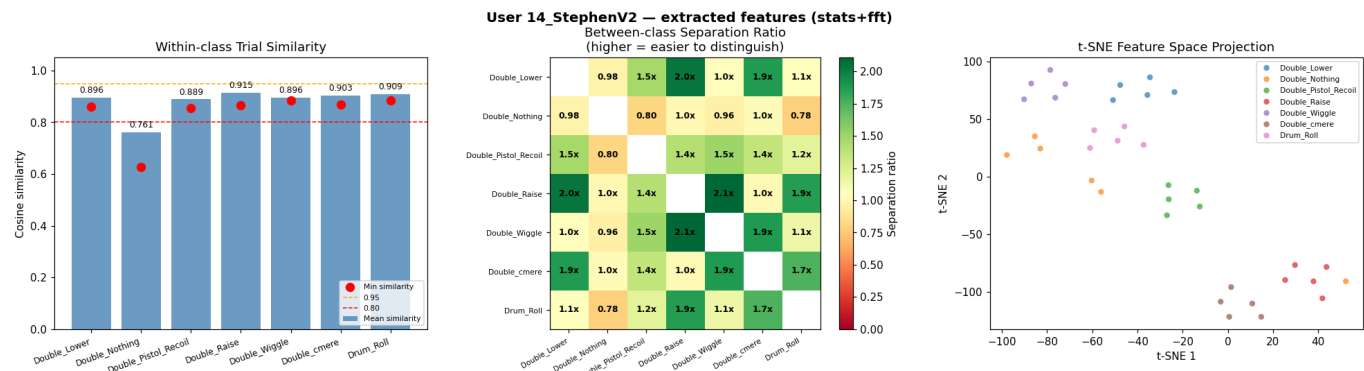
User 11_Mansh — extracted features (stats+fft)



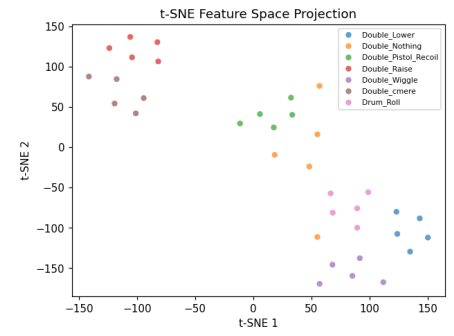
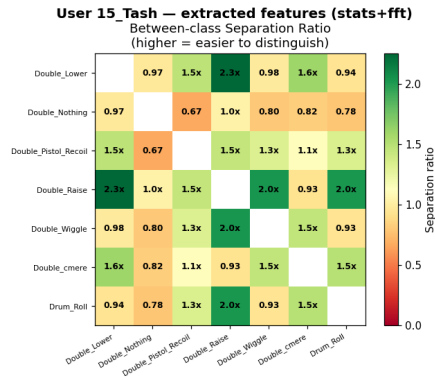
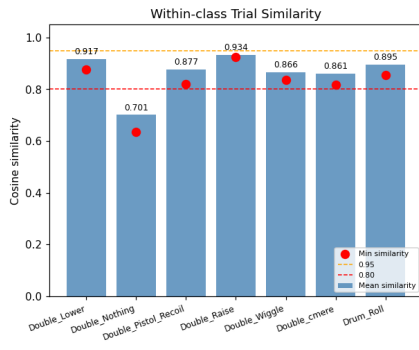
User 12_Marcus — extracted features (stats+fft)



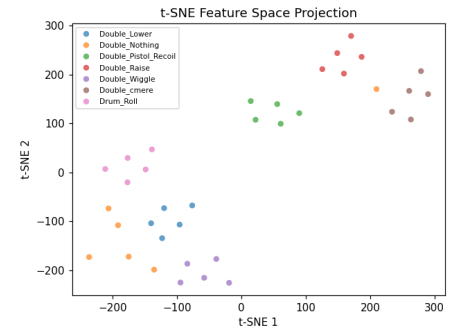
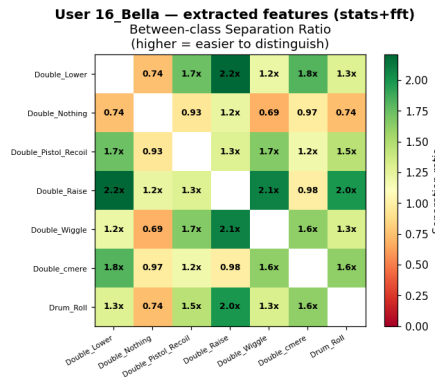
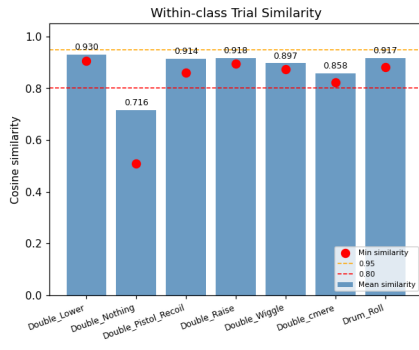
User 14_StephenV2 — extracted features (stats+fft)



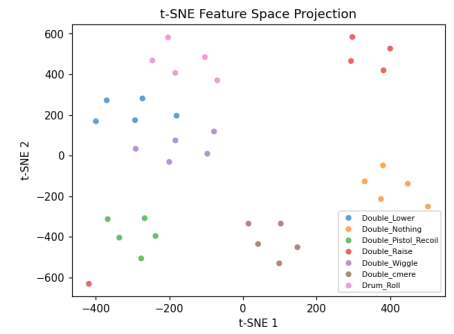
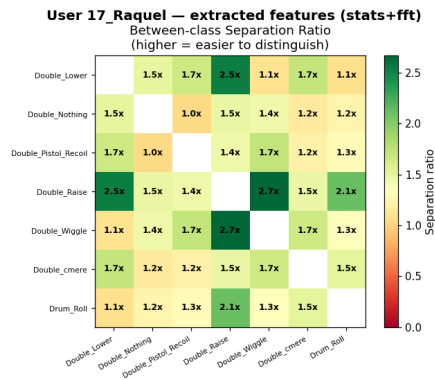
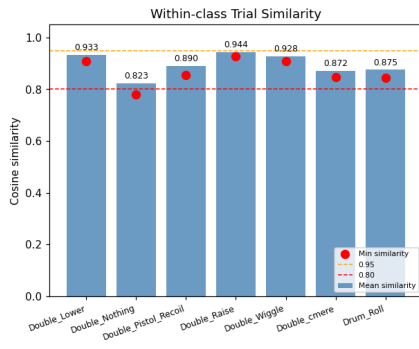
User 15_Tash — extracted features (stats+fft)



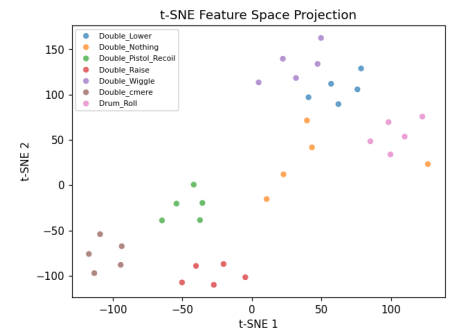
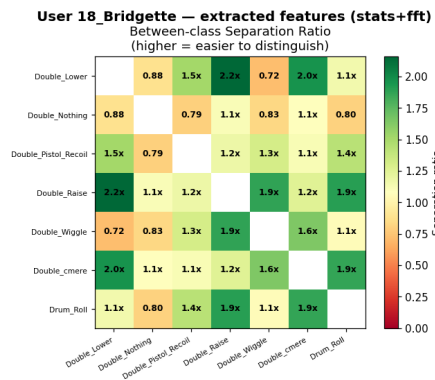
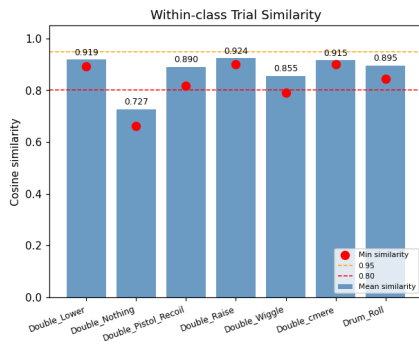
User 16_Bella — extracted features (stats+fft)



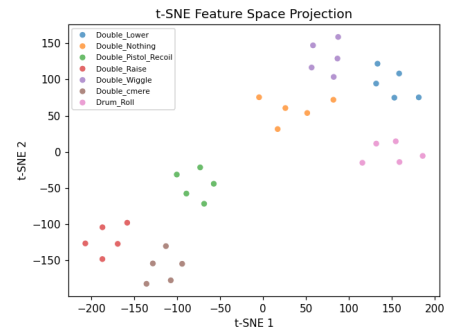
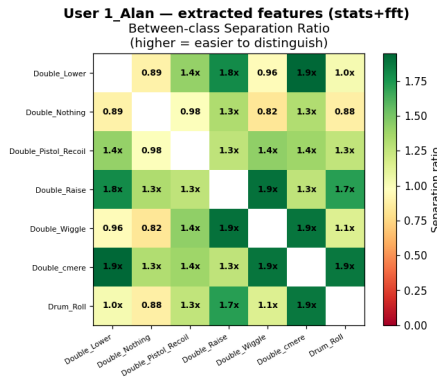
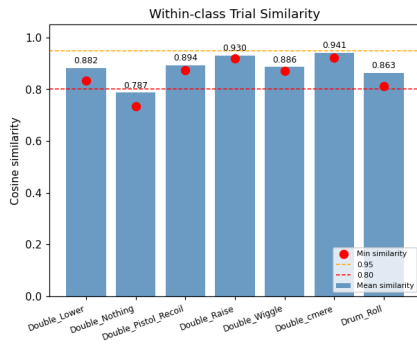
User 17_Raquel — extracted features (stats+fft)



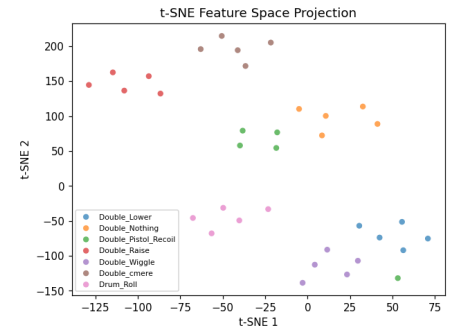
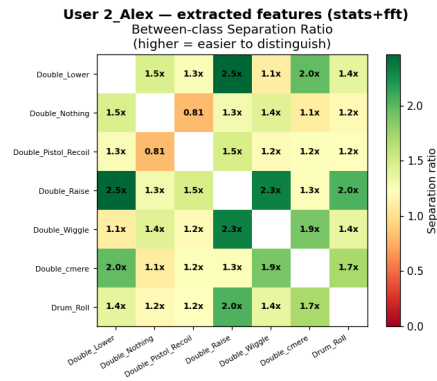
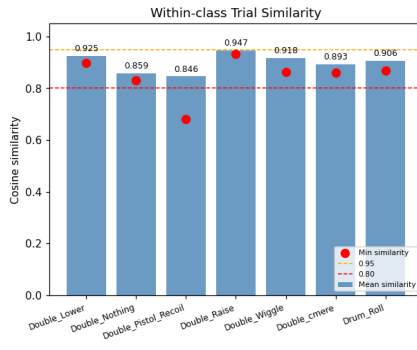
User 18_Bridgette — extracted features (stats+fft)



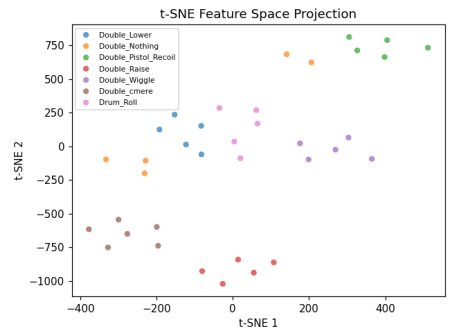
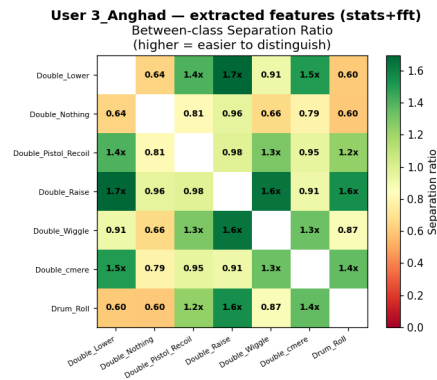
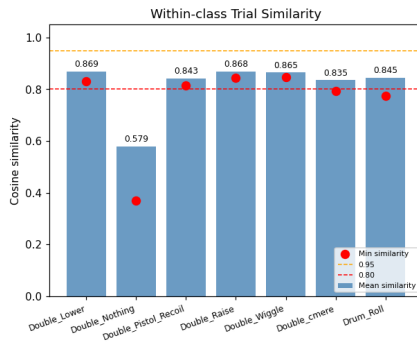
User 1_Alان — extracted features (stats+fft)



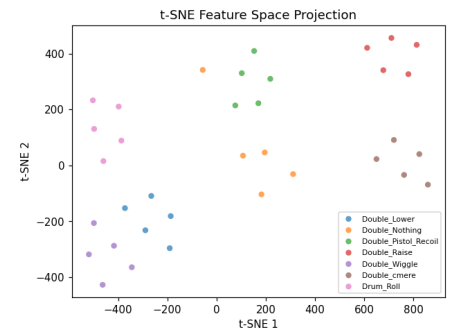
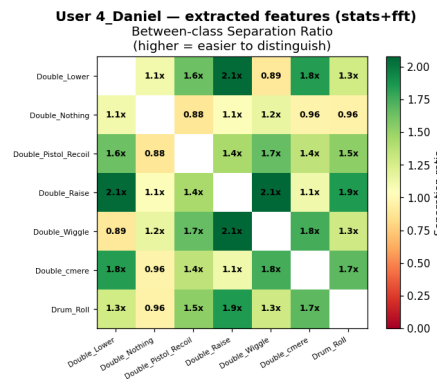
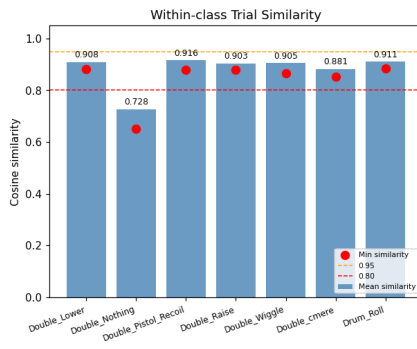
User 2_Alex — extracted features (stats+fft)



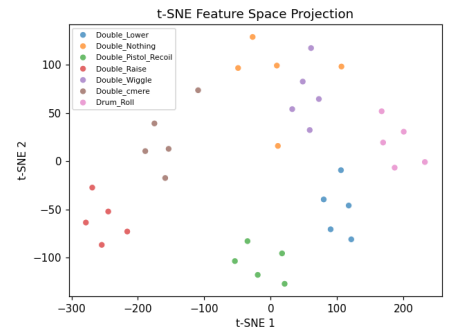
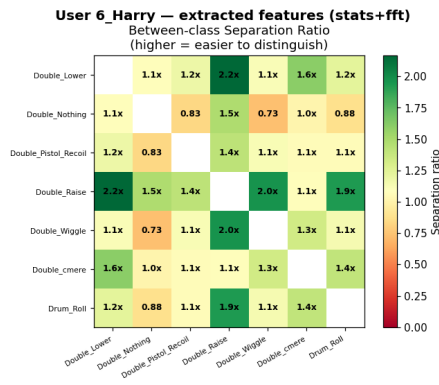
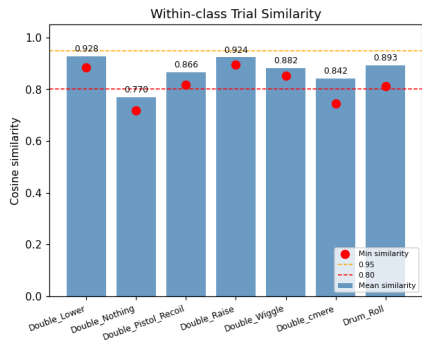
User 3_Angbad — extracted features (stats+fft)



User 4_Daniel — extracted features (stats+fft)



User 6_Harry — extracted features (stats+fft)



User 8_Jestin — extracted features (stats+fft)

